

Agile Software Development with Distributed Teams: Senior Management Support

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Abstract—Agile teams advocates face-to-face conversation, frequent communication and close collaboration between team members in order to work together as one team. In distributed software development, however, separation through space, time and culture affects the ability of the members of a distributed team to work together as one team. Through a Grounded Theory study that involved 55 participants from 38 different software companies in the USA, India and Australia, we investigate key concerns of distributed teams in Agile software development. We found that distributed teams need significant support from senior management in terms of organisational culture, human resource management, financial sponsorship, infrastructure and technology, and customer liaison. In this paper, we discuss the importance of senior management support for Agile software development with distributed teams.

Keywords-Senior Management, Agile Methods, Grounded Theory, Distributed Team, Global Software Development

I. INTRODUCTION

Team collocation is a hallmark of Agile software development [16], [41]. Collocation promotes face-to-face conversation, maximises informal communication and fosters close collaboration between team members. Agile teams carry out software development activities with the goal to provide business value to customer through early and continuous delivery of valuable software [30]. In Agile software development with distributed teams, however, the separation through space, time and culture affects the ability of the members of a distributed team to work together as one team [48], [50], [54].

Through a Grounded Theory study that involved 55 participants from 38 different software companies in the USA, India, and Australia, we investigate key concerns of distributed teams in Agile software development. We found that distributed teams need significant support from senior management in terms of organisational culture, human resource management, financial sponsorship, infrastructure and technology, and customer liaison.

The rest of the paper is structured as follows: section II presents a background on Agile software development, section III presents the Grounded Theory research method, section IV presents the results of the study, section V presents related work, section VI presents the limitation of the study, and finally section VII presents the conclusion.

II. BACKGROUND

In February 2001, seventeen independent software development practitioners gathered to recognise the core values of several “easy to follow” methods and wrote the Agile Manifesto [10]:

We are uncovering better ways of developing software by doing it and helping others do it.
Through this work we have come to value:

Individuals and interactions over process and tools
Working software over comprehensive documentation
Customer collaboration over contract negotiation
Responding to change over following a plan

That is, while there is value in the items on the right, we value the items on the left more.

These four core values emphasize the importance of frequent interaction between individuals in a team, continuous delivery of valuable software to customer, close collaboration between development team and customer, and prompt response to change in the project, particularly to provide business value to the customer. The 12 principles behind the Agile Manifesto [10] guide project teams to carry out software development activities while still upholding the core values.

Agile methods refer to a family of software development methods that follow an iterative and incremental style of development [1]. Agile methods include XP (eXtreme Programming) [8], Scrum [66], ASD (Adaptive Software Development) [42], DSDM (Dynamic Systems Development Method) [69], and FDD (Feature Driven Development) [60]. Scrum and XP are considered to be the most widely adopted Agile methods in software development projects [29]. Scrum mainly provides techniques for project management whereas XP focuses on software development practices.

Agile software development gives importance to the talents and skills of individuals and molds process to specific people and teams [16]. In particular, communication, collaboration and coordination between developers, customers and managers are considered to be important for the success of a project [58]. Beck [9] suggests that a project team in Agile software development comprises “a variety of people who work together in interlinking ways to make a project more effective”.

In this study, we define an Agile team as “a cross-functional, self-organizing, high performance team that carries out software development activities inline with the principles behind the Agile Manifesto in order to provide business value to customer”. A *cross-functional* team comprises everybody involved in the development of a project, including people from the business area, project management, development, testing, quality assurance and technical support [47]. In general, a cross-functional team as a whole has all required skills and knowledge to carry out a project. The members of a *self-organizing* team typically manage their workload, shift work among themselves based on need and best fit, and actively participate in team decision making while performing highly interdependent software development activities [41], [43]. In Agile software development, a self-organizing team captures the “spirit of Agile values and principles” through a focus on human and social aspects, trust and mutual respect [46]. A *high performance* team is a highly principle-driven, process-oriented, cohesive team which consistently focuses on the success of a project and constantly attempts to satisfy the customer [13], [52]. To summarise, an Agile team is a strategy to success even in complex software development projects.

A classic Agile team is a small, collocated team where the customer is an integral part of the team [43], [57]. In recent year, however, software companies are increasingly venturing into Agile software development with distributed teams. In this study, we define a distributed team as “a project team with members from the same organisation where at least one individual member is physically distant from the others in different countries, across different continents”. A distributed team faces challenges to work together as one team due to separation of team members through space, time and culture [48], [49], [53], [62]. Several studies [12], [28], [50], [56] point out that project teams require considerable support from senior management when integrating Agile methods in distributed software development. In this study, we investigate the importance of senior management support particularly for Agile software development with distributed teams.

III. RESEARCH METHOD

Grounded Theory (GT) is a systematic research method that emphasises the generation of theory derived from systematic and rigorous analysis of data. GT was jointly developed and first articulated by Barney G. Glaser and Anslem L. Strauss [40]. GT researchers gather data, particularly qualitative data from interviews and observations, and systematically discover a *theory* derived directly from the data [45], [57]. GT is intended neither to answer a specific research question nor to test an existing hypothesis, but rather to focus on discovery of a substantive theory regarding the research phenomenon within its context [4], [14], [61].

Grounded Theory developed into two versions – ‘Glaserian GT’ (classic GT) and ‘Straussian GT’ – due to the differences between Glaser and Strauss [14]. Glaser [35] argued that Straussian GT is no longer Grounded Theory that generates theories which are grounded in data, but rather “full conceptual

descriptions” that results from “forcing preconceived notions on data”. With the aim to generate a theory that is emergent from empirical data rather than from inferences or existing theories, we chose classic GT [33], [34], [35], [36], [37], [38], [39] as our research method for two main reasons. Firstly, GT is suitable to be used in areas that are under-explored or where a new perspective might be beneficial [2], [65], and literature on Agile software development with distributed teams is still scarce [26], [53]. Secondly, GT allows researchers to study social interactions and the behaviour of people [40], and Agile methods focus on people and their interactions in software development teams [66]. In the recent years, GT has been successfully used to study social aspects of Agile teams [31], [43], [57], [62], [73].

Glaser [40] suggests that a Grounded Theory study starts with a general area of interest rather than a specific research question because defining a research question before commencing a GT study leads to preconceived ideas or hypothesis of the research phenomenon [35], [36]. In this respect, we started this study with a general area of interest – Agile software development with distributed teams – rather than beginning with a specific research question. This does not mean that there is no specific problem for the research, but rather the research problem should belong to the subjects (individuals, groups or community) and thus should not be preconceived or forced [17], [36].

A. Research Context

This large-scale empirical research study involved 55 participants from 38 different software companies in the USA, India and Australia [19]. In order to get a rounded perspective of the research phenomenon, we interviewed participants from different roles in Agile software development: Developer, Quality Analyst (Application Tester), Business Analyst, Scrum Master, Agile Coach, Product Owner, and Senior Manager (e.g. Vice President of Delivery, Human Resource Manager and Director of Technology). Several participants were notably active in communities of practice where they have been participating in forums, authoring articles based on subject matter expertise and speaking at local events and international conferences. In this paper, we will only identify the participants using the codes P1 to P55 due to privacy and ethical consideration.

Scrum or a combination of Scrum and XP were used in the projects through adoption of fundamental practices such as small releases, continuous integration, estimating user stories, iteration planning, daily stand-up meetings, release planning, sprint reviews and sprint retrospectives. Team distribution varied between two and four locations in different countries across different continents, mainly North America and Asia. Project domain includes e-commerce, financial services, web applications and social networking. Project iteration varied between two and four weeks, and project duration varied between 6 and 36 months. Projects started with a small team size of 6 to 12 members but some teams had scaled through to a large team size in order to accommodate the increasing complexity of the projects.

B. Data Collection and Analysis

In GT, sampling occurs throughout the research study in order to facilitate the development of theory [40], [71]. The population under study is basically a set of concepts that constitute the phenomena rather than individuals experiencing the phenomena [2]. The initial decision for sampling is based on a general perspective and a general area of interest rather than preconceived theoretical ideas [11], [71]. After the emergence of the main concerns, sampling focuses on data collection for these main concerns, for the purpose of generating the conceptual hypothesis. Sampling in GT thus begins with *selective sampling* and moves into *theoretical sampling* when concepts begin to emerge.

Selective sampling is the identification of populations and settings prior to data collection. Theoretical sampling is the process of data collection directed by emerging theory rather than by predetermined population dimensions [25]. Whereas selective sampling is a *tentative* sampling strategy for beginning theory development, theoretical sampling is the *pivotal* sampling strategy in GT for developing a conceptually dense substantive theory. Glaser [33] describes that theoretical sampling occurs when the researcher jointly collects, codes and analyzes the data and decides what data to collect next and where to find them, in order to develop a theory as it emerges.

We conducted face-to-face, one-on-one interviews with participants using open-ended questions. With consent from participants, we voice-recorded interviews in order to concentrate on the conversation during the interviews. Initially a set of guiding questions was prepared to develop a smooth conversation with the participants. The questions mainly focused on the background of the project, the challenges faced by team members and the strategies that they adopt in order to overcome those challenges. We transcribed the interviews and analysed them using *key point open coding* to explore the meaning in the data [4]. Open coding breaks down, examines, compares, conceptualises and categorises the data [70]. We identified the key points from interview data and assigned a summary phrase, called a *code*, to each key point in order to produce a first level of abstraction. Using GT's *constant comparison* [34], we analysed each code with the codes from the same interview and those from other interviews. Codes related to a common theme were grouped together to produce a second level of abstraction called a *concept*. Constant comparison of the codes gave rise to many fresh concepts. We analysed the concepts using constant comparison, and concepts related to a common theme were grouped together to produce a third level of abstraction called a *category*.

Several key concerns emerged from the initial analysis: *senior management support*, *building trust*, *managing knowledge*, *promoting communication*, and *bridging cultural differences*. We continually adjusted our interview questions to focus on these key concerns. GT requires the theoretical sampling to be continued until *theoretical saturation* [40] is reached – that is when no more new concepts or categories emerge from the data and further data collection would be a waste of time.

We continuously wrote-up *theoretical memos* [36] on the ideas about the codes, concepts and categories as they emerge throughout data analysis. A theoretical memo is the “*the theorising write-up of ideas about substantive codes and their theoretically coded relationships as they emerge during coding, collecting and analyzing data*” [36]. Once theoretical saturation was reached, we began sorting the theoretical memos according to the conceptual ideation of the memos. *Theoretical sorting* of memos is an “*essential step*” for formulating the theory for presentation or writing because “*it begins to put the fractured data back together*” [33].

C. Emergence of Theory

The goal of GT is to generate a theory that accounts for a pattern of behaviour about what is going on in the area studied, which turns out to be the main concern of the participants and the research problem for the phenomenon being studied [33], [6], [55].

This research study has led us to discover a substantive grounded theory called “*The Theory of One Team*” which explains how a distributed team in Agile software development adopts explicit strategies for bridging spatial, temporal and socio-cultural distances, while facing critical impact factors, in order to work together as one team [19]. Different aspects of the theory has been described in different papers [20], [21], [22], [24], [23]. We found that *Senior Management Support* is an important factor which influences a distributed team to work together as one team. In this paper, we present the concepts *Organisational Culture*, *Human Resource Management*, *Financial Sponsorship*, *Infrastructure and Technology* and *Customer Liaison* which gave rise to the category *Senior Management Support*.

IV. RESULTS: SENIOR MANAGEMENT SUPPORT

In this section, we describe the importance of senior management support for distributed teams in Agile software development. Figure 1 shows the concepts *Organisational Culture*, *Human Resource Management*, *Financial Sponsorship*, *Infrastructure and Technology* and *Customer Liaison* that gave rise to the category *Senior Management Support*. Using diagrams to illustrate the emergence of a concept or a category is recommended in GT [3], [32].

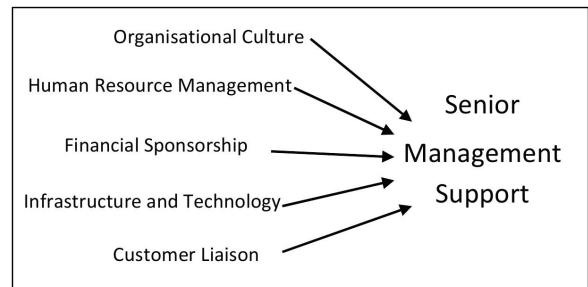


Fig. 1. Emergence of category *Senior Management Support*.

We have selected quotations drawn from our interviews that shed particular light on the emergent concepts. Due to space constraints, however, we cannot describe *all* the underlying key points and codes from our interviews that further ground the emergent concepts and categories.

A. Organisational Culture

Organisational culture has a significant impact on the social nature of Agile teams in the organisations. Agile teams can represent a major cultural change for senior management since Agile teams value responding to change, even late in development, over following a plan. Senior management, however, requires detailed plans and schedules and focuses on risks and opportunities in projects, in order to commit for supporting the teams. Success of Agile adoption in an organisation relies on the synergy of Agile teams and senior management through “leadership-and-collaboration” management:

“There is a need for change in mindset of [senior] management, towards leadership behaviours that encourage collaborative working. Agility needs individuals whom are natural leaders rather than managers. It needs a genuine commitment from senior management at the organisation to be a successful [Agile] team.” – P6, Developer.

Agile adoption in an organisation requires support from senior management particularly in the form of acceptance and recognition to the changes that comes with combining Agile methods with software development activities. Senior management needs to recognise changes in the structure and roles of the self-organising, cross-functional Agile teams, and provide support to the team to achieve a collective success in software development projects:

“Agile adoption needs a change in [organisational] culture. The whole team has to come together to do it, [and] you need to have the [support] from senior managers who play an important role in the organisation. I strongly believe that you have to start gaining acceptance at the management level, and then from team members, so that everyone feels that they have a stake in the success of the Agile [projects].” – P44, Product Owner.

A cohesive Agile team practices frequent interaction, effective communication and close collaboration between members of a team across global sites. Therefore, it is sensible for senior management to create a common culture across different sites of the team and treat everyone as equal as possible in order to foster a collaborative nature of a cohesive Agile team:

“Agile [adoption] needs a cultural change in the team and organisation. It’s really important for team members from offshore [location] to feel equal to the team members from onsite [location]. Management has to make them as co-equal as possible, as same as possible, and create a really good working environment for the team to enjoy working together.” – P3, Scrum Master.

A culture established through cooperative social practices between Agile teams and senior management that understands and recognises Agile values and principles is critical for the success of Agile adoption in organisations:

“Senior management was building Agile culture within the [organisation] and transforming our organisation. Management understands the value of Agile and provides 100% support to the teams. The success in Agile [adoption] within the team is incredible. You need support and buy-in from senior management, [otherwise] Agile adoption will eventually fizzle out in the organisation.” – P46, Agile Coach.

Senior management in an organisation stabilises and reinforces organisational culture over time with regulations and policies for work practices and everyday activities of individuals in the organisation. Since Agile methods rely substantially on interaction, collaboration and communication, the key for successful Agile adoption in an organisation is the team. Therefore, senior management needs to create a culture that fosters Agile adoption within the team and organisation.

B. Human Resource Management

Participants suggested that senior management must learn to manage Agile teams differently than traditional software development teams, particularly to resolve human resource issues such as team-based versus individual rewards, skills requirements and team building activities. While a cross-functional Agile team as a whole has all required skills and every team member is willing to communicate and collaborate for building the software, it is important for the senior management not to review and reward them for individual performance:

“Senior management needs to change the way Agile teams are managed. For example, senior management should consider how are individuals being reviewed and rewarded, or what programmes can be put in place to ensure that the team comes together, or what training do individuals in the team need so that the team is going to be successful.” – P46, Agile Coach.

Since interaction and communication are considered key tenets of Agile teams, senior management should give particular importance to the hiring criteria in the organisation. Ill-fitting individuals who do not work well together, often due to the lack of soft skills, may destroy the collaborative nature of Agile teams. Therefore, senior management is also responsible for creating team chemistry through hiring mechanisms:

“We give a lot of importance to soft skills when we hire people. We are particular about hiring people who are social and able to communicate. There are lots of technical people who are brilliant developers but are not able to work collaboratively with others. We realise that working on an Agile project can be a very socially intense experience.” – P20, Senior Management.

Participants suggested that senior management often standardise skillset available within a team at different global sites through relevant training sessions, particularly on the technologies that are used in the projects. Hiring professional trainers and experienced consultants, and sending teams for training sessions and certifications can be expensive and time consuming. Therefore, senior management support is essential to standardise practices, design training programmes and successfully implement them across the organisation:

“We have a schedule for training programmes that everybody in the organisation needs to attend, particularly a set of soft-skills courses such as personal-development, communication, management training, and leadership. And, from the technical side, we train them on [software] development processes, Agile practices, and what we do in the project.” – P45, Senior Management.

Though individuals bring with them adequate skillset to perform the job, ongoing support to upgrade existing skills or to acquire new skills can be useful for the individuals. Formal training programmes that include soft-skills, technical skills and Agile practices have been found useful to upgrade skills of team members. Managing human resource in terms of hiring right individuals to fit Agile teams, rewarding individuals for team performance, and upgrading skills of team members are important responsibilities of senior management.

C. Financial Sponsorship

Agile teams need financial sponsorship from the organisation for software development activities, such as acquiring reliable hardware and software, purchasing information resources and attending training courses for acquiring knowledge and travelling to visit team members at different locations. Senior management support in terms of financial sponsorship is important for successful Agile adoption in an organisation:

“We provide good infrastructure and a lot of training to prepare them for [software] development projects. The [financial] cost for these [activities] are considered high for our small company [but] we still provide financial support for them.” – P29, Senior Management.

Learning is an important aspect of successful Agile teams. Through giving continuous attention to technical skills and soft skills, Agile teams quest for different knowledge at different time from various knowledge sources. Financial sponsorship for attending conferences and training courses can be useful for encouraging team members to gain new knowledge from experts or communities of practice:

“Our company has policies for attending conferences and certification training courses. We just have to ask for financial sponsorship to attend them and if the senior managers find them useful and offers good value, they are willing to pay for that kind of courses.” – P55, Developer.

Realising the importance of face-to-face interaction and working together in close proximity, senior management ensure that whenever a new team is formed, one side of the team travels to the other location and works collocated for a short duration before distributing the teams. Some teams organise cross-location visits and rotation between team members in different locations for building team interaction and trust. These practices can be expensive, but senior management that understands the importance of such practices provide financial sponsorship to the team members:

“At the beginning of a project, it’s important for the entire team and the customer to be collocated in order to facilitate communication between the development team and the customer. Though this practice can be expensive, it’s recognised as good practice.” – P20, Senior Management.

Senior management that supports distributed teams is willing to provide financial sponsorship for encouraging continuous learning, travelling across different locations of the teams, and acquiring required hardware and software for gaining competitive advantage in Agile software development.

D. Infrastructure and Technology

Collocated Agile teams communicate through face-to-face conversation, and share information on project status on walls of the team-room in the form of product backlog list, sprint backlog list, burndown chart, or story cards. Conversely, distributed teams use technology-mediated communication to carry out even basic communication activities and share project-related information between individuals in the team. Therefore, support from senior management in terms of investing in suitable infrastructure and technology for communication, collaboration, or knowledge sharing is essential for distributed teams:

“Tools can make it easier and effective to share information with people in remote locations. Most information sharing doesn’t happen during the meetings, [rather] it happens incidentally during one-on-one phone calls or instant messaging.” – P20, Senior Management.

Combination of different tools are particularly important for distributed teams to conserve and distribute knowledge gathered by team members in different locations. Individuals in the team can find different tools are suitable for communication on different situations. Therefore, availability of several different tools can be appropriate for promoting communication and managing knowledge in distributed teams:

“We use video conferencing, and [other] tools for communication [such as] Skype, Microsoft Share View, WebEx, Messenger, and Google Documents. We have a mix of tools because we know that we can’t always use one single tool for communication. That’s a good technological investment for our company.” – P55, Developer.

Participants discuss that senior management understands the importance of using technology for supporting distributed teams, and therefore often provides them with reliable infrastructure and required technology such as high-end computers with useful communication and collaboration tools to support team members in different locations:

“At all times, I provide all the support that the teams need from my capacity as the Vice President of Delivery in this organisation. They have really good laptops, all the required tools for communication and knowledge sharing, and I also enrol them for online training [in order] to gain knowledge on [Agile] software development practices.” – P28, Senior Management.

Since distributed team members need to communicate and collaborate using technology-mediated tools, senior management should invest in reliable infrastructure and useful technology that promotes communication and collaboration in the team. Considerable amount of information is readily available on the Internet, and therefore additional support from senior management in the form of knowledge management tools that promote the use of information sources and integrating them into development process can be beneficial for distributed team members.

E. Customer Liaison

The Standish Group CHAOS Report [51] highlights the importance of customer involvement and senior management support in software development projects. Realising the importance of customer collaboration, Agile teams advocate that customers and developers must work together on a daily basis throughout the project. Senior management that understands this particular principle, strive to provide active support for Agile teams in terms of customer liaison:

“At any time there has been very active support for client management. We bring along customers to work with the team [in order] for them to understand the way things are going, and we solicit new practice so that we can work together better.” – P48, Senior Management.

Though Agile methods require presence of onsite customers or customer representatives working together with development teams, finding a customer who is willing to be involved and available for the teams can be difficult. Senior management can influence customers to play a significant role in the projects through frequent interaction with development team and providing them with timely feedback:

“It has been really hard to catch up with the customer and [therefore] we don’t ever feel that we are part of a team [instead] we feel like a contractor. In this case, [senior] management has helped to bridge the gap between the teams in different places and the customer. We receive a lot of great support from senior management.” – P55, Developer.

In traditional software development, customers and developers do not engage in direct communication because business analysts are often responsible to draft out project requirements and customers do not see the software until project completion. This can lead to miscommunications between the customers and developers. To be successful in Agile software development with distributed teams, senior management should request customers for a certain level of collaboration and time commitment from them:

“The expectations were not set properly. The customer would throw a lot of work onto the team and expect the team to deliver the work and understand the whole vision. But, the team didn’t exactly get the whole picture of the vision. So, the VP of Delivery decided that we flew the customer in with the team for a week. That was a huge support [from senior management].” – P52, Scrum Master.

Besides individuals in the project teams, Agile adoption in software development also affects customers where significant level of commitments in terms of time and effort are expected from them. Close collaboration between customers and project teams can (a) ensure project requirements are clearly understood, (b) provide development team with early feedback on acceptance tests, and (c) lead to project success.

V. DISCUSSION

In this section, we discuss our findings in the light of existing literature. In this study, we sought out to discover the main concerns which our participants continually resolve while carrying out software development activities. When research investigates “what” and “how” human participants do what they do then statistical techniques are often used to inform us of the probability of the reliability of those results. When research, however, attempts to find explanations of “why” human participants do what they do then quantitative counting of how many times things happen is insufficient to answer the research questions and we have to turn to qualitative methods. For this very reason, the techniques of Grounded Theory (GT) were “discovered” rather than invented in the 1960’s [40]. GT finds the meaning in the data, discovers what the data was telling us, not in numbers but the real meaning underlying the conversations between people.

Using GT method, we discovered “The Theory of One Team” [19] which explains how a distributed team in Agile software development adopts explicit strategies for bridging spatial, temporal and socio-cultural distances, while facing critical impact factors, in order to work together as one team. Though this study was not specifically an investigation into whether senior management support is important for distributed teams or how distributed teams could secure support from senior management support, we discovered that senior management support for distributed teams was a main concern for our participants. Further in-depth investigation has led us to discover how senior management support influences the members of a distributed team to work together as one team.

The importance of senior management support for co-located Agile teams has been widely acknowledged [15], [27], [44]. Further, several studies [28], [62] have pointed out the importance of senior management support particularly for Agile software development with distributed teams.

Ramesh et al. [62] investigated challenges that arise from blending agility with distributed development and how those challenges can be addressed. They studied three organisations that adopted Agile methods in distributed software development and identified several practices to address challenges to communication, control and trust across distributed teams. They report that significant support from senior management in terms of improving communication, facilitating knowledge sharing and building trust demonstrates a balance between Agile adoption and distributed software development.

Livermore [56] reports that senior management support for software development teams has a strong impact on Agile adoption in organisations. He found a significant correlation between the success of Agile adoption and senior management support. He suggests that senior management should ideally provide Agile teams with support in terms of access to useful resources and training on software development methods.

Through an experience report, Farmer [28] reflects on the success of Agile adoption in a large, distributed team. She highlights that among the major factors that contributed to the success of Agile adoption was the timely support from senior management, especially in the area of customer liaison and autonomy in decision making within the team.

Rottier and Rodrigues [63] discuss experience of the software development team in a medical device company. They report that substantial support from senior management has led to the success of Agile adoption within the software development team. Senior management encouraged other individuals in the company to attend the daily Scrum meetings and sprint review meetings, and developed a shared understanding of Agile software development within the company.

In this study, we found that distributed teams need support in terms of organisational culture, human resource management, financial sponsorship, infrastructure and technology, and customer liaison. We suggest that a distributed team must proactively seek support from senior management and the senior management must consciously provide continual support to distributed teams in order for the team members to work together as one team despite the physical separation through space, time and culture between them.

VI. EVALUATION OF THE FINDINGS

There is much dialogue centred on the difficulty of establishing validity criteria in qualitative research [5], [7], [67], [68], [72]. Creswell [18] outlined “specific criteria” that must be employed in any study for validating qualitative research: triangulation, member checking, audit trail, disconfirming evidence, prolonged engagement, rich description, self-reflection, collaboration and peer debriefing. Interpretive researchers, however, rule out methodology as the basis for validity and

argue that establishing “specific criteria” continues the positivist assumption of an underlying objective reality to which research results can be compared and judged for truth [5], [64]. Due to the considerable debate about what it means to do valid research in the field of qualitative inquiry, reaching the desired goal and meeting the requirement of trustworthiness become particularly problematic for interpretive, qualitative researchers [5], [59]. Angen [5] suggests that interpretive, qualitative research should consider validation from the *ethical* and *substantive* perspectives. Angen [5] argues that validation is an ethical question that must be addressed from inception to completion of research – when research is formulated, carried out and written up. A substantive approach to validation indicates researchers need to document the chain of interpretations in order for others to judge the trustworthiness of the meanings arrived at in the end [5].

This interpretive, qualitative research study adopts the *ethical* and *substantive* validation approaches of good qualitative research, and then creatively combines the theoretical sensitivity of the researcher with GT’s rigorous analysis process to produce valuable interpretations. To achieve the ethical validation, we have diligently followed the processes in GT from the start until the end of the research study. For example, we avoided a huge upfront literature review, we used theoretical sampling for data collection and analysis until we reached theoretical saturation, we continually wrote theoretical memos on the code, concept and categories, and we performed theoretical sorting in order to develop a grounded theory on the research phenomena. To achieve the substantive validation, we included selected interview quotations throughout the results section in order to document evidence of how the theory was developed in this research study.

Glaser [33], [35] suggests *credibility* of a grounded theory can be evaluated through four criteria: *fit*, *work*, *relevance* and *modifiability*. *Fit* refers to “the ability of the categories and their properties to fit the realities under study in the eyes of the subjects, practitioners, and researchers in the area” [36]. In order to *fit*, the categories and their properties, which define the theory, must have naturally emerged from the data and were not forced by preconceived ideas or hypothesis [35]. *Work* refers to “the ability of the theory to explain the major variations in behaviour in the area with respect to the processing of the main concerns of the subjects” [36]. In order to *work*, a grounded theory must be able to explain what happens in the data, predict what will happen, and interpret what is happening in the area studied, for both participants and researchers [35]. *Relevance* is achieved when a grounded theory “allows core problems and processes to emerge” [33]. In order to be *relevant*, participants and practitioners must consider that the theory is useful, meaningful and applicable to the people working in the substantive area [35]. *Modifiability* refers to “the quality of the theory to be ready for changes to include variations in emergent properties and categories caused by new data” [36]. In order to be *modifiable*, a grounded theory must be ready for changes to include variations in emergent categories and properties caused by new data [35].

Fit is considered the most important criterion for evaluating the credibility of a theory, whereas the criteria *work*, *relevance*, and *modifiability* supports the “fitness” of the theory. Urquhart et al. [71] argue theoretical sampling is the single most important contributor to the “fit” of a theory and without theoretical sampling, it will be impossible to establish how “saturated” the theory is. In this study, theoretical sampling guided us in the selection of suitable participants who provided explanations on the emerging main concerns and the evolution of interview questions that focused on the emerging concerns. We interviewed 55 participants from *different* roles in the Agile software development, across 38 *different* software companies in three *different* countries, namely the USA, India and Australia. We also continually changed the interview questions in order to focus on the concerns that seem central to the emerging theory. We continued theoretical sampling until we noticed complete redundancy, or theoretical saturation, in the new data. We do not claim that our findings are universally generalisable to all distributed Agile teams, but rather our findings accurately characterize the contexts studied.

VII. CONCLUSION

Through a Grounded Theory study, we investigate the main concerns of distributed teams in Agile software development. Over a period of three years, the study involved 55 participants from 38 different software companies in the USA, India and Australia. While several main concerns have been discussed in different papers, in this paper we report that senior management must provide distributed teams with significant support in terms of organisational culture, human resource management, financial sponsorship, infrastructure and technology, and customer liaison in order for the team members to work together as one team despite the physical separation through space, time and culture between them.

We hope that findings from this study can provide senior management with insight on how to support distributed teams in Agile software development. We are mindful that there can be other form of useful support that a distributed team may need from senior management but did not emerge from our analysis. Our study, however, raises some interesting avenues for future research. First, this study can be replicated within a different context, such as involving an outsourced team with members from different organisations. The findings that will emerge from the replicated study can be compared against our existing findings. In this way, researchers can generate a *formal theory* to describe the importance of senior management support for Agile teams in global software development in general. Another avenue for future study is to measure the findings from this study using quantitative research method to validate the emergent of concepts and category. For example, a survey can be carried out to investigate how many teams gain the different forms of support from senior management as presented in this paper.

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